

Fig. 4: PoE depends on two devices at each end of the cable—the power sourcing equipment (PSE) device and the powered device (PD)

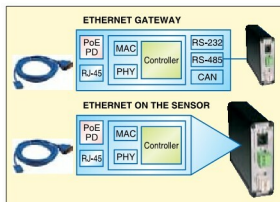


Fig. 5: Basic sensors are expected to use Fieldbus connected to a gateway with Ethernet as the backbone

industrial environments where high temperatures and destructive transient ESD events often occur. Fortunately, PoE devices designed to survive under these conditions are available in the market.

Other benefits of PoE include the size reduction that can be realised by removing AC power converters on the equipment. As mentioned earlier, installation costs can be reduced as use of multiple power cables and conduits can be avoided, increasing the ease of installation.

When interconnecting equipment across different industrial network standards, extensive vendor support is often required. However, using a standard network with standardised power opens the door for product competition, which improves support availability and equipment

diversification, ultimately reducing the overall cost. The PoE IEEE 802.3af and the preliminary 802.3at standards are in place to ensure compliance between devices receiving PoE.

Evaluating these business drivers demonstrates how PoE brings substantial value to industrial applications. The next question is: Where will this new technology penetrate the industrial network, and what are the challenges?

Bringing value to industrial networks

Today, industrial Ethernet is penetrating the industrial network core but has not yet migrated to the edge of the network. It started with the enterprise network and reached the controller level. Below that level, Fieldbus is the most common communication standard. Small sensor devices, such as pushbuttons, valve control blocks, and other control or sensing devices are connected via Fieldbus as shown in Fig. 1. So where does PoE fit in and how does it add value?

Ethernet can connect devices currently communicating via Fieldbus. Complex sensors such as level measurement or optical sensors, HMI and



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